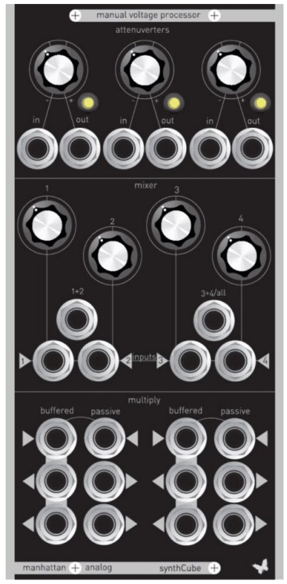
Manual Voltage Processor (MVP) for DotCom   Brian Dunn 241002

The MVP was designed back in 2012 in Eurorack format by Jason Coates (Manhattan Analog). Featuring three attenuverters, a mixer section that can be a dual 2 input or single quad input, and four multiples either buffered or passive it is simply a great utility module.

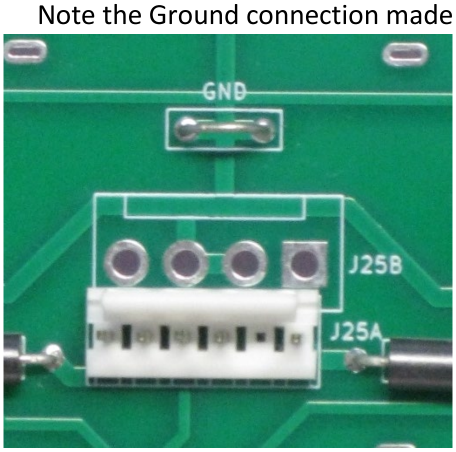


Original Eurorack MVP   Caleb Condit's DotCom   Dunn redesign

Shortly thereafter, Caleb Condit designed a DotCom front panel but the Eurorack PCBs couldn't easily be adapted to the DotCom components and was difficult to build. I did build two of them and wrote a how-to manual for it.
In 2024, I decided to develop a DotCom version from the ground up. The design is a single PCB, no wires.

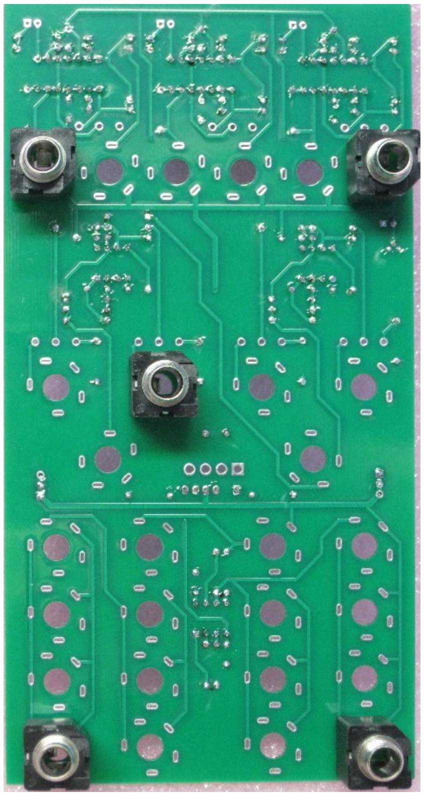
Recommended Tools:
Solding Station & Solder
½" socket wrench (for Jacks)
10mm socket wrench (for pots)
Couple of pencils or rods to support/lift the assembly or help align jacks and pots.

Assembly Instructions:
1) Populate all the PCB components except the jacks, pots, LEDs, and ICs.
    Note the IC socket orientations.
    Note the Ground connection made from 24 gauge wire or component lead



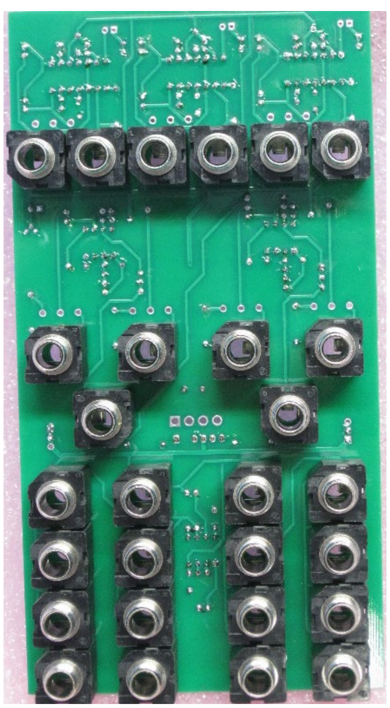
2) This is a good time to check for power supply shorts or even apply power and check the IC sockets and 5V regulator.

3) Use the pencils to raise the PCB. Place (do not solder yet) five jacks in the corners and center of the PCB. Lower the Front Panel over the jacks and install the jack washers and nuts finger tight.



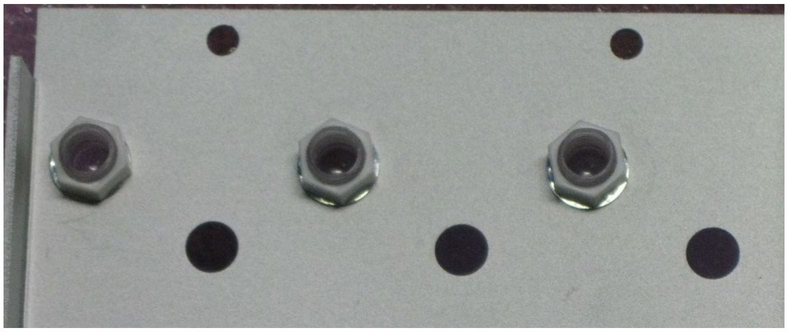
5) Flip the “sandwich” over and solder the jacks. Press down on the PCB if necessary to make sure the jacks are fully seated against the PCB and solder the jacks.

6) Remove the nuts and washers then, likewise, place the rest of the jacks in the PCB. Place the panel onto the jacks (I found it easiest to tilt the panel onto the jacks). Install the same washers and nuts as before. Flip and solder all the jacks as before.



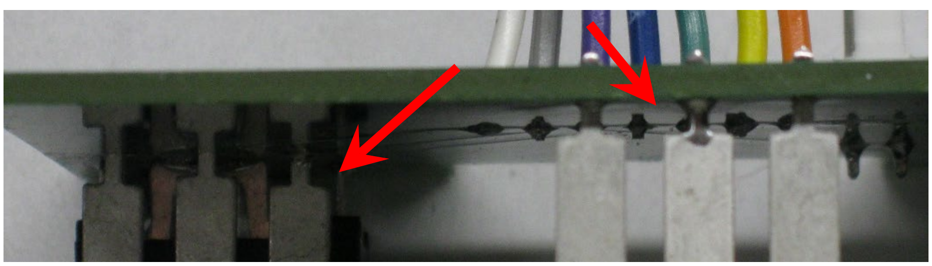
7) Remove the nuts and washers.

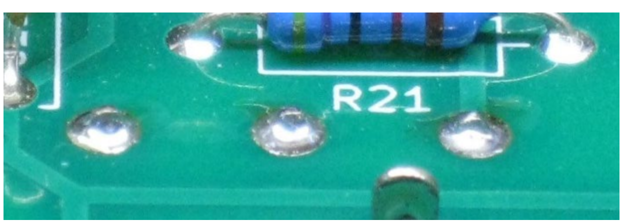
8) Install the LED lenses into the panel. Start them with your fingers; then use the 10mm socket.



9) If the pots have dust caps, install them now. Note the caps are shaped to fit around the lead opening.

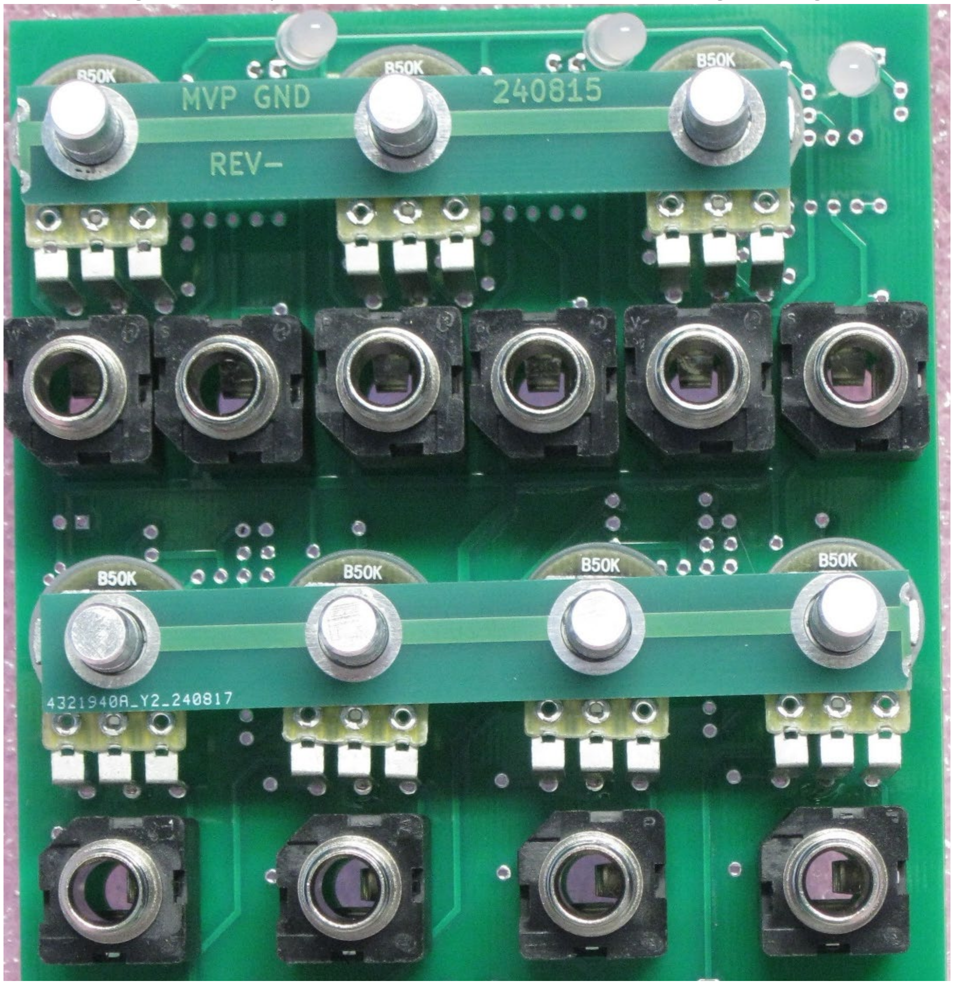
10) NOTE: When the unit is completely assembled, there will be a gap between the pot legs' notch and the PCB. The pot tips will just clear the top of the PCB.





Place the pot leads into the PCB holes.
Tack solder one leg such that it just comes out of the PCB and that the legs are perpendicular to the PCB. You can do this on the top side of the PCB.
Add the pot spacers. Note that the thin side of the spacers are towards the top of the PCB.

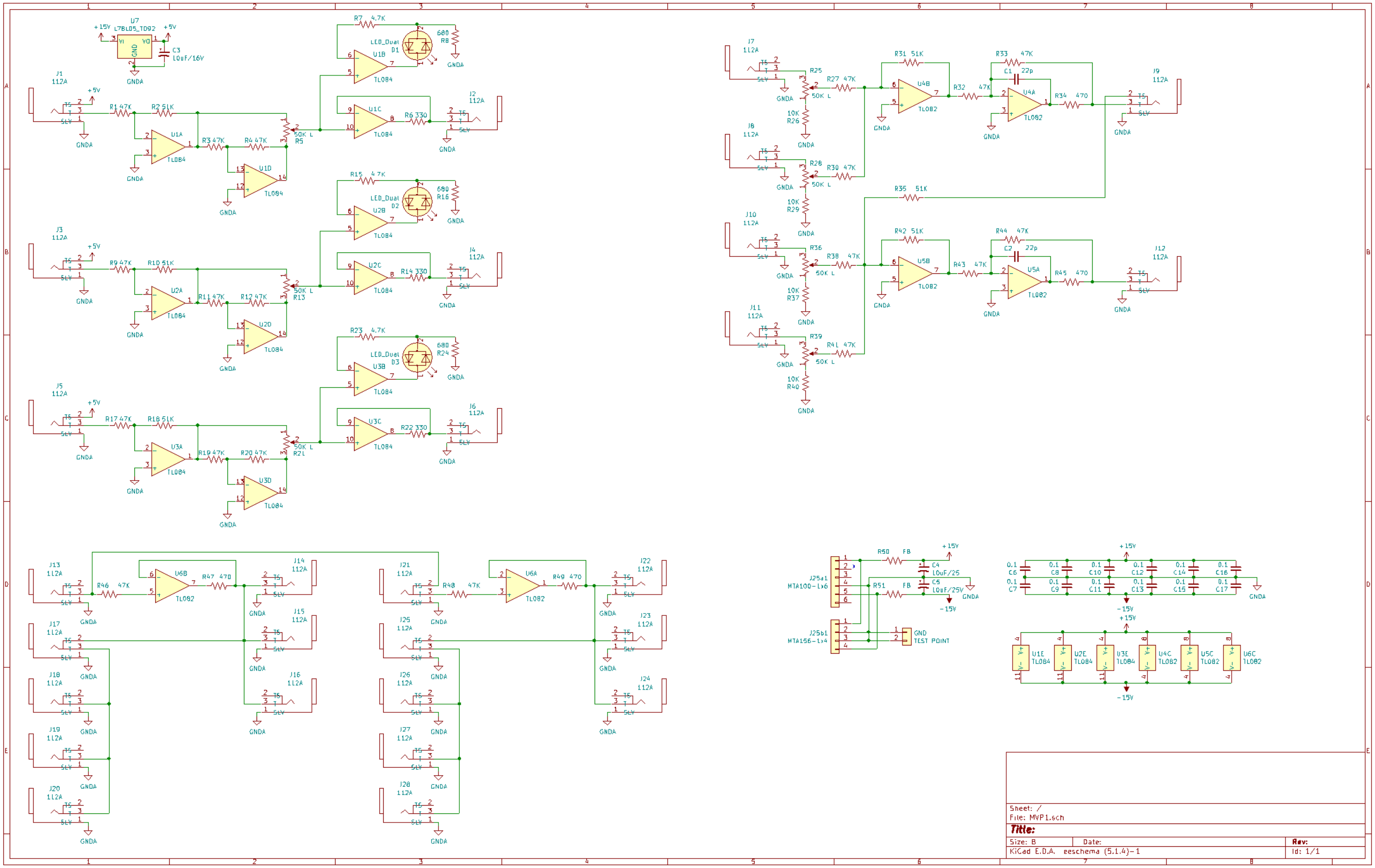
11) ADD THE LEDS NOW but do NOT solder.
Short lead goes into square hole for Red Positive/ Green Negative signals.



12) The spacers let the pot leads float in the PCB holes to get the proper spacing noted above.
Carefully place the panel over the pot shafts and jacks.
You will probably have to gently align the pots to start them into their holes but do not force the assembly together.
Once the pots are aligned and the jacks enter their holes, install the washers and nuts to catch a couple turns.
For the jacks, just use the four corner jacks in the picture above.
Likewise the four outside corner pots will lift the other pots in that row.
Keep checking that the pot leads are not getting bent and tighten the nuts in a pattern.
Once the jacks and pots nuts are fully seated, add all the other jack and nut washers and nuts.
Reflow the previously tacked pot leads (They may “snap” into place.).
Solder the rest of the pot leads.

13) With the panel facing down, the LEDs should drop into their lenses. Make sure they are even and solder them.

14) Install the ICs and knobs.



Sheet: /
File: MVP1.sch
Title:
Size: B
Date:
Rev:
KiCad E.D.A. eeschema (5.1.4)-1
Id: 1/1